IntelDoc KG Demo — Handoff Brief

Nov 4, 2026 · @Hemang

Build a demo for a law-enforcement audience: ~5,000 synthetic PUBS.XML intelligence cables about fictional Mexican cartels (2010–2026), a knowledge graph extracted from them with ground truth to score against, and a desktop app where a single search box answers questions with citations. Documents are rendered with Haiku via the API; the world is built first so every cable is sampled from known facts. Budget: ~$30 of API spend. Nothing here is real: every cartel, person, official and company is invented; only the geography is real.

1. Demo story — what the audience sees

An analyst-style desktop app with one search box. The presenter types a plain-English question, gets an answer with inline citations to specific cables and paragraphs, and clicks through to the entity's neighborhood in the graph and its timeline. Then the presenter switches persona from a Secret//NOFORN user to an Unclassified//FOUO user, asks the same question, and the answer changes because the graph carries portion marks on every fact. The closing slide shows the extraction scorecard: how much of the known ground-truth graph the pipeline recovered from 5,000 cables, measured, not claimed.

The audience should leave believing three things: the ingest understands real IC document structure (PUBS.XML, ISM markings, NTK), the graph answers questions keyword search cannot (multi-hop, temporal, cross-agency), and the whole thing can be measured because the corpus has ground truth.

2. Non-negotiables

- **Everything named is fictional.** Cartels, factions, people, nicknames, officials, companies, vessels, phone numbers, accounts. Real geography only: Mexican states, cities, ports, border crossings, highways; US border cities and distribution hubs. Keep a blocklist of real Mexican criminal organizations and real named officials (seed it from public reference lists); the checker rejects any cable that contains one. The world bible ships with the demo so anyone in the room can verify it is invented.
- **World first, cables second.** No cable is written free-form. Every cable is sampled from the world's events and entities, so the corpus has a complete ground-truth graph before extraction runs.

- **The LLM writes only narrative paragraphs.** All identifiers, dates, markings, NTK profiles, publisher blocks, keywords and coverage are rendered deterministically from the record model.
- **Classification roll-up is computed, never templated.** Paragraph portion marks are generated first; `PreformattedText`, `DocumentBody` and the root inherit the maximum classification and the union of dissemination controls.
- **Resumable everywhere.** A SQLite manifest tracks each cable's state (planned, drafted, critiqued, revised, validated, failed). Re-running the generator never regenerates a validated cable. Batch job IDs are persisted so a session timeout costs nothing.
- **LLM provider is a config value.** `anthropic` (default: Haiku 5.5 for rendering and extraction, Opus 5.5 for the world bible and QA grading) and `ollama` (optional, for an air-gapped query-time story). Same interface, same prompts.
- **Hard spend cap.** `MAX_SPEND_USD` enforced by a token accountant that counts every call; the run stops when it is reached. Default cap: $40.
- **Nothing in the repo is sensitive.** The PUBS.XML structure is from the public ODNI release; the content is invented. No real document text, no real markings copied from real documents.

4. Phase 0 — World bible

The world is a JSON/YAML dataset under `world/`, built in two steps: a hand-reviewed seed, then a scripted expansion.

Seed (`world/seed.yaml`, written with Hemang in chat, not generated unattended). One page: 5 cartels/factions with a 16-year arc each (founding or split, 2–4 leadership changes, at least one fracture, one alliance that later breaks), the 20–30 plazas/corridors they fight over, and the 30–40 biggest timeline beats (major arrests, extraditions, assassinations, named LE operations, elections that move corrupt officials in or out).

Expansion (`scripts/expand_world.py`, Opus 5.5, structured output, iterative). Grows the seed into the full world and validates it with code (no dangling references, no overlapping leadership of one org, no event dated after a participant's death or arrest unless the event is the release). Target sizing:

Entity type	Count	Notes
Cartels / factions	5–6	Each with `validFrom` / `validTo` membership and leadership edges
People	150–300	Tiers: boss, lieutenant, plaza boss, sicario chief, launderer, logistics, lawyer, corrupt official (police, customs, mayor, prosecutor, military). 2–4 aliases each; some nicknames shared between two people; some aliases inherited when a boss falls
Plazas / corridors	20–30	Control changes hands over time; each a real place or pair of places
Front companies	60–100	Trucking, produce export, currency exchange, used cars, mining, hospitality; some owned through nominees
Vessels, aircraft, vehicles	40+	Semi-submersibles, pangas, light aircraft, tractor-trailers with plate numbers
Phones, accounts	300–500	Rotated over time; some shared across people (the alias-resolution trap)
Events	400–600	Typed: Shipment, Seizure, Arrest, Extradition, Killing, Meeting, Payment/Bribe, Operation, Designation, Election, Split, Alliance. Each with date, place, participants with roles, quantities where relevant
Sources	40–60	Fictional source IDs per agency with a reliability grade that is stable over time

Every entity has a canonical name, a stable `id`, alias list, and the real places it is tied to. The world is the ground truth for Phase 5 evaluation: `world/graph_truth.json` is derived from it by script and never edited by hand.

5. Phase 1 — Record model and cable sampling

A **cable record** is a Pydantic model holding everything the XML needs except the narrative text. The sampler (`scripts/plan_cables.py`) produces N records from the world and writes them to the manifest before any LLM call.

Per record: `cable_id`; agency and office; document type (IIR for most, plus a few summary/assessment products); report date; 1–3 events covered, with the subset of participants and facts the reporting agency plausibly knows; the source (from the world's source list) and its reliability grade; classification plan (S//NF or U//FOUO at the document level, with a per-paragraph mark plan); coverage place names; keywords; identifiers; NTK profile; and the **planted-facts list** — the exact entities (by canonical name or a chosen alias), dates, quantities and relationships the narrative must contain. The planted-facts list is the ground truth for that cable.

Corpus mix (parameters in `config/corpus.yaml`):

Dimension	Target
Agency	DEA 55%, FBI 12%, HSI/CBP 12%, DIA 6%, Treasury/OFAC-style 5%, OSINT press summaries 10%
Classification	S//NF 65%, U//FOUO 35%; OSINT always U
Relevance	70–80% core (about the world's events); 20–30% marginal or noise (routine port activity, unrelated smuggling, background on a place)
Contradictions	~5% of cables carry a fact that disagrees with another cable (a different date, a different count, a source that misidentifies someone)
Corrections	1–2% are follow-up cables that correct or recall an earlier cable by DocumentID, linked in `Relation`
Alias usage	Each person referred to by a nickname or partial name in at least 40% of mentions
Time	Uniform over 2010–2026 with a bump around each major timeline beat; the same event appears in 2–6 cables from different agencies over days to weeks
Length	3–8 numbered paragraphs, 250–900 words of body

Sampling is seeded (`--seed`) so a corpus is reproducible from the world and the config.

6. Phase 2 — Renderer, narrative generation, checker

Three stages per cable, all driven from the record: generate the narrative, validate it, render the XML.

6.1 Narrative generation (Haiku 5.5). Input: the planted-facts list, the agency/source persona, the classification plan, and 2–3 exemplar cables (hand-written, in `exemplars/`). Output: numbered paragraphs, each opening with its portion mark, plus the SUBJ line. Three passes: *draft* (facts to paragraphs in IIR register), *critique* (a second call checks the draft against the facts and a style rubric and lists defects), *revise* (applies the critique). Temperature moderate on draft, low on revise. Optional `best_of: 2`. Paragraphs must mention every planted fact, must not introduce proper nouns outside the world bible, and must keep the agency's partial view (an FBI cable does not know what only the DEA source knows).

6.2 Deterministic checker (no LLM). Fails the cable back to regeneration on any of: a planted entity missing (canonical name or the alias the record chose); a capitalized multi-word token that matches neither the world bible, the real-geography gazetteer, nor an allowlist of common words; a blocklisted real name; a date or quantity that differs from the record; a paragraph without a portion mark or with a mark higher than the cable's plan; numbering not 1..N; length outside bounds. Failures are logged with the reason so the prompts can be tuned from the pilot.

6.3 XML rendering. Jinja2 templates from the corrected skeleton below, `lxml` for well-formedness, and an XSD/Schematron hook (`validate/`) that runs if a PUBS.XML schema package is dropped into `schemas/` (the public ODNI V2018 package or whichever version the target ingest uses). Two attribute sets, chosen by document classification:

Attribute	Secret cable	Unclassified cable
`ism:classification`	`S`	`U`
`ism:ownerProducer`	`USA`	`USA`
`ism:disseminationControls`	`NF`	`FOUO` (or omitted)
`ism:classificationReason`	`1.4(c)`	omitted
`ism:declassDate`	createDate + 25 years	omitted

Attribute	Secret cable	Unclassified cable
`ism:declassException`	`25X1`	omitted
`ism:derivedFrom`, `ism:derivativelyClassifiedBy`	`Unspecified`	omitted
`ism:createDate`, `ism:compliesWith="ICD-710"`	present	present
`ism:resourceElement="true"`, `DESVersion`s	root only	root only

Roll-up: each paragraph's mark is from the plan; `PreformattedText`, `DocumentBody` and `IntelDoc` get the maximum classification and the union of dissemination controls across paragraphs. Never cross S with FOUO or U with NF.

Corrected skeleton (whitespace and tag errors from the hand-typed template fixed; `cia:` extension element spellings kept as in the source feed; `[...]` = rendered from the record):

```
<pubs:IntelDoc xmlns:pubs="urn:us:gov:ic:pubs:relaxed"
xmlns:ism="urn:us:gov:ic:ism"
    xmlns:irm="urn:us:gov:ic:irm" xmlns:ntk="urn:us:gov:ic:ntk"
xmlns:cia="urn:us:gov:cia"
    xmlns:xlink="http://www.w3.org/1999/xlink"
    COIProtectionLevel="3" DESVersion="5" irm:DESVersion="3"
ism:DESVersion="5" ntk:DESVersion="3"
    ism:resourceElement="true" ism:createDate="[YYYY-MM-DD]"
ism:compliesWith="ICD-710"
    [S or U attribute set from the table above]>

  <ntk:Access ism:classification="U" ism:ownerProducer="USA">
    <ntk:AccessProfileList ism:classification="U" ism:ownerProducer="USA">
      <ntk:AccessProfile ism:classification="U" ism:ownerProducer="USA">
        <ntk:AccessSystemName ism:classification="U"
ism:ownerProducer="USA">IC_COI4</ntk:AccessSystemName>
        <ntk:AccessProfileValue ism:classification="U"
ism:ownerProducer="USA" ntk:vocabulary="IC_COI4_DISSEMTO">[agency]
</ntk:AccessProfileValue>
        <ntk:AccessProfileValue ism:classification="U"
ism:ownerProducer="USA"
ntk:vocabulary="IC_COI4_ISSUE">CN</ntk:AccessProfileValue>
```

```
      </ntk:AccessProfile>
      <ntk:AccessProfile ism:classification="U" ism:ownerProducer="USA">
        <ntk:AccessSystemName ism:classification="U"
ism:ownerProducer="USA">POLICY-URI</ntk:AccessSystemName>
        <ntk:AccessProfileValue ism:classification="U"
ism:ownerProducer="USA" ntk:vocabulary="URI">policy://[nnnn]/[n-n]
</ntk:AccessProfileValue>
      </ntk:AccessProfile>
    </ntk:AccessProfileList>
  </ntk:Access>

  <irm:NoticeList [doc-level ism attrs]>
    <irm:Notice irm:nonISMNoticeType="BANNER" [doc-level ism attrs]>
      <irm:NoticeText [doc-level ism attrs]>WARNING: [handling text]
</irm:NoticeText>
    </irm:Notice>
  </irm:NoticeList>

  <pubs:PublicationMetadataList>
    <pubs:PublicationMetadata>
      <pubs:AdministrativeMetadata>
        <pubs:IdentifierList>
          <pubs:DocumentID>[DEA-NN-IIR-NNNN-YY]</pubs:DocumentID>
          <pubs:InternalID internalIDtype="IngestionID">[9 digits]
</pubs:InternalID>
          <pubs:InternalID internalIDtype="GUIDE">guide://[NNNN]/[xxxxxx]
</pubs:InternalID>
          <pubs:InternalID internalIDtype="messagingSystemID">[YY]-
HQS[digits]</pubs:InternalID>
        </pubs:IdentifierList>
        <pubs:Publisher>
          <pubs:AgencyAcronym acronym="[DEA]" vocabulary="Nautilus"/>
          <pubs:AuthorInfo><pubs:OfficeName>[office]</pubs:OfficeName>
</pubs:AuthorInfo>
          <pubs:POCInfo><pubs:OfficeName>[agency]</pubs:OfficeName>
<pubs:EmailAddress/></pubs:POCInfo>
          <pubs:AttributionText [ism attrs]>[attribution]
</pubs:AttributionText>
        </pubs:Publisher>
        <pubs:DateList>
          <pubs:DatePublished>[YYYY-MM-DDThh:mm:ssZ]</pubs:DatePublished>
          <pubs:ExtensionElements><cia:DateReceived>[YYYY-MM-DDThh:mm:ssZ]
</cia:DateReceived></pubs:ExtensionElements>
```

```
        </pubs:DateList>
        <pubs:Rights>
          <pubs:PrivacyActIndicator indicator="false"/>
          <pubs:CopyrightActIndicator indicator="false"/>
        </pubs:Rights>
        <pubs:Relation>
          <pubs:AlternateFormat id="entities"
xlink:href="https://nautilus.example.ic.gov/resource/[uuid]/entities"
              xlink:role="https://www.iana.org/assignments/link-
relation/alternate" xlink:title="entities"/>
          <pubs:AlternateFormat
xlink:href="https://nautilus.example.ic.gov/resource/[uuid]/[id].orig"
              xlink:role="https://www.iana.org/assignments/link-
relation/alternate" xlink:title="[id].orig"/>
        </pubs:Relation>
        <irm:ProductionMetricsList [ism attrs]>
          <irm:ProductionMetric [ism attrs]>
            <irm:ProductionMetricsSubject [ism
attrs]>Unspecified</irm:ProductionMetricsSubject>
            <irm:ProductionMetricsCoverage [ism
attrs]>Unspecified</irm:ProductionMetricsCoverage>
          </irm:ProductionMetric>
        </irm:ProductionMetricsList>
        <pubs:OtherProperty
qualifier="DocumentPrecedence">0</pubs:OtherProperty>
        <pubs:OtherProperty qualifier="HEADER">[routing header]
</pubs:OtherProperty>
        <pubs:ExtensionElements>
          <cia:AdminstrativeMetadataExtensions>
            <cia:DocumentStatus>ACTIVE</cia:DocumentStatus>
            <cia:OrigintingSystem>LEGACY</cia:OrigintingSystem>
          </cia:AdminstrativeMetadataExtensions>
        </pubs:ExtensionElements>
      </pubs:AdministrativeMetadata>
      <pubs:DescriptiveMetadata>
        <pubs:Title [ism attrs]>SUBJ: ([mark]) [subject line]</pubs:Title>
        <pubs:Language encoding="ISO639-3">eng</pubs:Language>
        <pubs:Description [ism attrs]>None</pubs:Description>
        <pubs:Coverage>
          <pubs:GeospatialExtent>
            <pubs:GeographicIdentifier>
              <pubs:PlaceName [ism attrs]>[place]</pubs:PlaceName>
            </pubs:GeographicIdentifier>
```

```
          </pubs:GeospatialExtent>
        </pubs:Coverage>
        <pubs:Subject><pubs:Keyword>[keyword]</pubs:Keyword></pubs:Subject>
        <pubs:IntelType><pubs:IntelDiscipline>[HUMINT|OSINT|SIGINT]
</pubs:IntelDiscipline></pubs:IntelType>
        <pubs:ProductLine qualifier="true">Unspecified</pubs:ProductLine>
        <pubs:CollectSourceList>
          <pubs:CollectSource>
            <pubs:CollectSourceDescription>
              <pubs:Para [ism attrs]>SOURCE: [description]  CONTEXT:
[reliability]</pubs:Para>
            </pubs:CollectSourceDescription>
          </pubs:CollectSource>
        </pubs:CollectSourceList>
        <pubs:ExtensionElements>
          <cia:DescriptiveMetaDataExtensions>
<cia:Collection>CC</cia:Collection></cia:DescriptiveMetaDataExtensions>
        </pubs:ExtensionElements>
      </pubs:DescriptiveMetadata>
    </pubs:PublicationMetadata>
  </pubs:PublicationMetadataList>

  <pubs:DocumentBody [rolled-up ism attrs]>
    <pubs:PreformattedText [rolled-up ism attrs]>
SUBJ: ([mark]) [subject line]

1. ([mark]) [paragraph]
2. ([mark]) [paragraph]
...
TOPIC: [codes]
REQUIREMENT: [codes]
DATE OF ACQUISITION: [date]
POC: [office]
WARNING: [handling]
======================== DISSEMINATION ================================
AGENCY: [list]   U.S. MISSION: [list]   MILITARY: [list]   STATE/LOCAL:
[list]
DECLASSIFY ON: [date or event]
END OF MESSAGE [mark]
    </pubs:PreformattedText>
  </pubs:DocumentBody>
</pubs:IntelDoc>
```

Filenames follow `NNNN-xxxxxx.xml`; the GUIDE, filename, AlternateFormat UUID, DocumentID year, messagingSystemID prefix and `createDate` are all derived from one record so they never disagree. Output to `corpus/xml/`, with the per-cable planted-facts and paragraph marks to `corpus/truth/` as JSON.

Known uncertainties (confirm against a real header if one is available; otherwise keep the template's values): the IRM and NTK DESVersion numbers paired with PUBS V5; whether `COIProtectionLevel` is allowed by the target XSD; whether the second NTK profile's `policy://` value sits under the `URI` vocabulary (as rendered here) or elsewhere.